Problem · Unit 3 · 10 points · 60 min · High difficulty / trap

P11 — Pendulum whose string is hauled through a ring: a rheonomic constraint, angular momentum and the work of the hauling force

A bob P of mass m hangs from a light inextensible string that passes through a small, smooth ring O fixed to the ceiling. The other end of the string leaves the ring and is hauled in so that the length between O and P is

$$\ell(t) = \ell_0 - u\,t, \qquad u > 0 \text{ constant}$$

The motion is planar, in the vertical plane of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ (x horizontal, y vertical upward, uniform gravity $-g\,\mathbf{j}$, $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ out of the page). The angle θ is measured from the downward vertical, positive towards $+x$, and the string stays taut. Use the unit vectors $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ (from O to P) and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$.

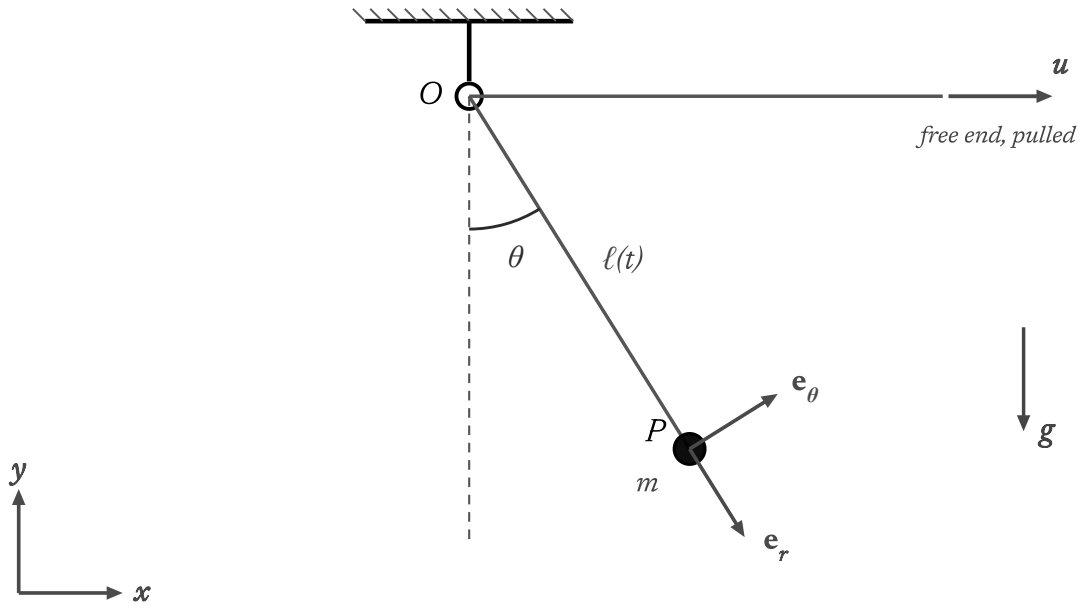


Figure 1. Pendulum hauled through a fixed smooth ring O: length $\ell(t) = \ell_0 - ut$ between O and the bob P, angle θ from the downward vertical (positive towards $+x$), unit vectors $\mathbf{e}_r$ (from O to P) and $\mathbf{e}_\theta$; the free end is pulled with constant speed u.

Setup

- Data: m, g, ℓ_0, u. Numerical parts: (d) and (e) $m = 0.5$ kg, $\ell_0 = 2$ m, $u = 0.5$ m s^{-1}, $g = 0$, initial transverse speed $\ell_0\dot{\theta}_0 = 1$ m s^{-1}; (f) $\ell_0 = 1$ m, $u = 0.01$ m s^{-1}, $g = 9.81$ m s^{-2}, $\theta(0) = 0.05$ rad, $\dot{\theta}(0) = 0$.
- Hypotheses: point particle, massless inextensible string, smooth ring (the tension T is the same on both sides), no air drag, S_0 inertial, $\ell > 0$.

Part (a) · 1 point

Degrees of freedom and the constraint

How many degrees of freedom has P? Classify the constraint imposed by the string.

Step 1. Count

In the vertical plane a free particle has 2 degrees of freedom. The taut string between O and P imposes

$$f(\mathbf{r}, t) = |\mathbf{r}| - (\ell_0 - ut) = 0$$

one equation, so 1 degree of freedom remains: θ. The length is **prescribed** by whoever pulls the string, so it is not a second coordinate (Notes §2.7).

Step 2. Classification

The constraint involves only positions and time: it is holonomic. Time appears explicitly, so it is **rheonomic**: the constraint moves, and its reaction (the tension) can do work on P, unlike the fixed string of the ordinary pendulum. A string can only pull, so the constraint is unilateral and holds while $T \geq 0$.

KEY RESULTS

- $|\overrightarrow{OP}| - \ell_0 + ut = 0$: holonomic, **rheonomic** (depends on t), and unilateral ($T \geq 0$).
- $2 - 1 = 1$ degree of freedom, coordinate $\theta(t)$; $\ell(t)$ is a given function, not an unknown.

Part (b) · 1.5 points

Kinematics in polar coordinates

Write the position, velocity and acceleration of P in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$, first for a general length $\ell(t)$ and then for $\ell = \ell_0 - ut$.

Step 1. Change of basis and unit vectors

The working basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ turns with P; its relation to the inertial basis is

$$\begin{pmatrix}\mathbf{e}_r\\ \mathbf{e}_\theta\\ \mathbf{k}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{i}\\ \mathbf{j}\\ \mathbf{k}\end{pmatrix}, \qquad [R] = \begin{pmatrix}\sin\theta & -\cos\theta & 0\\ \cos\theta & \sin\theta & 0\\ 0 & 0 & 1\end{pmatrix}$$

Checks: rows 1 and 2 have length $\sin^2\theta + \cos^2\theta = 1$ and $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \sin\theta\cdot\sin\theta - (-\cos\theta)\cos\theta = +1$, so the basis is right-handed ($\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$). By the chain rule, $\dot{\mathbf{e}}_r = \dfrac{d\mathbf{e}_r}{d\theta}\dot\theta = (\cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j})\dot\theta = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = (-\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j})\dot\theta = -\dot\theta\,\mathbf{e}_r$.

Step 2. Position and velocity

$\mathbf{r} = \ell\,\mathbf{e}_r$. Product rule:

$$\mathbf{v} = \dot\ell\,\mathbf{e}_r + \ell\,\dot{\mathbf{e}}_r = \dot\ell\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta$$

Step 3. Acceleration

Differentiating each product:

$$\mathbf{a} = \ddot\ell\,\mathbf{e}_r + \dot\ell\dot\theta\,\mathbf{e}_\theta + \big(\dot\ell\dot\theta + \ell\ddot\theta\big)\mathbf{e}_\theta + \ell\dot\theta\,\dot{\mathbf{e}}_\theta = \big(\ddot\ell - \ell\dot\theta^2\big)\mathbf{e}_r + \big(\ell\ddot\theta + 2\dot\ell\dot\theta\big)\mathbf{e}_\theta$$

Step 4. Prescribed length

$\ell = \ell_0 - ut$ gives $\dot\ell = -u$, $\ddot\ell = 0$:

$$\mathbf{v} = -u\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a} = -\ell\dot\theta^2\,\mathbf{e}_r + \big(\ell\ddot\theta - 2u\dot\theta\big)\mathbf{e}_\theta$$

KEY RESULTS

- $\mathbf{v} = \dot\ell\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta,\ \mathbf{a} = (\ddot\ell - \ell\dot\theta^2)\,\mathbf{e}_r + (\ell\ddot\theta + 2\dot\ell\dot\theta)\,\mathbf{e}_\theta.$
- With $\dot\ell = -u, \ddot\ell = 0$: $\mathbf{v} = -u\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta,\ \mathbf{a} = -\ell\dot\theta^2\,\mathbf{e}_r + (\ell\ddot\theta - 2u\dot\theta)\,\mathbf{e}_\theta.$

Part (c) · 2 points

Forces and equations of motion

Write the forces on P and obtain the equation of motion free of the tension and the tension itself.

Step 1. Forces

Weight: $\mathbf{W} = -mg\,\mathbf{j}$ with $\mathbf{j}\cdot\mathbf{e}_r = -\cos\theta$ and $\mathbf{j}\cdot\mathbf{e}_\theta = \sin\theta$, so $\mathbf{W} = mg\cos\theta\,\mathbf{e}_r - mg\sin\theta\,\mathbf{e}_\theta$. Tension: the string pulls P towards the ring, $\mathbf{T} = -T\,\mathbf{e}_r$, $T \geq 0$.

Step 2. Transverse projection

$$m\big(\ell\ddot\theta + 2\dot\ell\dot\theta\big) = -mg\sin\theta \implies \ell\ddot\theta + 2\dot\ell\dot\theta + g\sin\theta = 0$$

With $\dot\ell = -u$: $(\ell_0 - ut)\ddot\theta - 2u\dot\theta + g\sin\theta = 0$. The term $-2u\dot\theta$ has the form of a **negative damping**: hauling the string in feeds the swing.

Step 3. Radial projection

$$m\big(\ddot\ell - \ell\dot\theta^2\big) = mg\cos\theta - T \implies T = m\big(g\cos\theta + \ell\dot\theta^2 - \ddot\ell\big) = m\big(g\cos\theta + \ell\dot\theta^2\big)$$

because $\ddot\ell = 0$: a constant hauling speed adds no radial inertia term. The equation of Step 2 is also what a Lagrange equation (extension, not in the course notes) with $L = \tfrac12 m(\dot\ell^2 + \ell^2\dot\theta^2) + mg\ell\cos\theta$ gives.

KEY RESULTS

- $\mathbf{W} = mg(\cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta)$, $\mathbf{T} = -T\,\mathbf{e}_r$.
- $\ell\ddot\theta + 2\dot\ell\dot\theta + g\sin\theta = 0$, i.e. $(\ell_0 - ut)\ddot\theta - 2u\dot\theta + g\sin\theta = 0$.
- $T = m\big(g\cos\theta + \ell\dot\theta^2 - \ddot\ell\big) = m\big(g\cos\theta + \ell\dot\theta^2\big)$.

Part (d) · 1.5 points

Angular momentum about O and the gravity-free case

Compute $\mathbf{H}_O$ and $d\mathbf{H}_O/dt$. Is it conserved? Then remove gravity (a tether in orbit reeled in at constant speed, $g = 0$): find $T(\ell)$ and, for the data of the Setup, $\dot\theta$, the speed and T when $\ell = 0.5$ m.

Step 1. Angular momentum

$\mathbf{H}_O = \mathbf{r}\times m\mathbf{v} = \ell\,\mathbf{e}_r \times m\big(\dot\ell\,\mathbf{e}_r + \ell\dot\theta\,\mathbf{e}_\theta\big) = m\ell^2\dot\theta\,(\mathbf{e}_r\times\mathbf{e}_\theta)$, and $\mathbf{e}_r\times\mathbf{e}_\theta = \mathbf{k}$ (it is the cross product of $\mathbf{i}$ and $\mathbf{j}$ rotated by $\theta - 90^\circ$):

$$\mathbf{H}_O = m\ell^2\dot\theta\,\mathbf{k}$$

Step 2. Its rate (Notes §4.8)

The tension passes through O, so its moment is zero; the weight gives $\mathbf{r}\times\mathbf{W} = \ell\,\mathbf{e}_r \times (-mg\sin\theta\,\mathbf{e}_\theta) = -mg\ell\sin\theta\,\mathbf{k}$. Product rule:

$$\frac{d}{dt}\left(m\ell^2\dot\theta\right) = m\left(2\ell\dot\ell\dot\theta + \ell^2\ddot\theta\right) = m\ell\left(\ell\ddot\theta + 2\dot\ell\dot\theta\right) = -mg\ell\sin\theta$$

which is the equation of (c) multiplied by $m\ell$. With gravity $\mathbf{H}_O$ is not conserved.

Step 3. Gravity-free case

With $g = 0$ the right-hand side vanishes: $\ell^2\dot\theta = h$ constant. In the tension, $\dot\theta = h/\ell^2$:

$$T = m\ell\dot\theta^2 = m\ell\frac{h^2}{\ell^4} = \frac{mh^2}{\ell^3}$$

The transverse speed $\ell\dot\theta = h/\ell$ grows as the string shortens, and the tension grows like ℓ^{-3}: whoever reels the tether in must pull harder and harder.

Step 4. Numbers

$h = \ell_0\cdot(\ell_0\dot\theta_0) = 2\cdot 1 = 2\ \mathrm{m^2\,s^{-1}}$. $\ell = 0.5$ m is reached at $t = (2-0.5)/0.5 = 3$ s. Then $\dot\theta = 2/0.25 = 8\ \mathrm{rad\,s^{-1}}$, $\ell\dot\theta = 4\ \mathrm{m\,s^{-1}}$, $|\mathbf{v}| = \sqrt{u^2 + (\ell\dot\theta)^2} = \sqrt{0.25+16} = 4.031\ \mathrm{m\,s^{-1}}$ and $T = 0.5\cdot 4/0.125 = 16$ N, compared with $T = 0.5\cdot 4/8 = 0.25$ N at the start.

KEY RESULTS

- $\mathbf{H}_O = m\ell^2\dot\theta\,\mathbf{k}$; $\dfrac{d\mathbf{H}_O}{dt} = \mathbf{r}\times\mathbf{W} = -mg\ell\sin\theta\,\mathbf{k}$: **not** conserved.
- $g = 0$: $\ell^2\dot\theta = h$, $T = m\ell\dot\theta^2 = mh^2/\ell^3$, which grows without bound as $\ell \to 0$.
- Data: $h = 2\ \mathrm{m^2\,s^{-1}}$; at $\ell = 0.5$ m ($t = 3$ s): $\dot\theta = 8\ \mathrm{rad\,s^{-1}}$, $\ell\dot\theta = 4\ \mathrm{m\,s^{-1}}$, $|\mathbf{v}| = 4.031\ \mathrm{m\,s^{-1}}$, $T = 16$ N (it was 0.25 N at ℓ_0).

Part (e) · 2 points

Who supplies the kinetic energy?

Show that $\dfrac{d}{dt}\big(E_{kin}+V\big)=T\,u$ *and explain why this is the power of the hauling force. For* $g=0$ *and the data of (d), compute the work done by the hauling force between* $\ell_0=2$ *m and* $\ell=0.5$ *m and check it against the change in kinetic energy.*

Step 1. Power of the tension

$\mathbf{T}\cdot\mathbf{v}=-T\,\mathbf{e}_r\cdot\big(\dot\ell\,\mathbf{e}_r+\ell\dot\theta\,\mathbf{e}_\theta\big)=-T\dot\ell=Tu$. On a fixed string ($\dot\ell=0$) the tension is orthogonal to the velocity and does no work; here the constraint moves and it does.

Step 2. Energy balance (Notes §4.4)

The work-energy theorem gives $\dfrac{dE_{kin}}{dt}=\mathbf{W}\cdot\mathbf{v}+\mathbf{T}\cdot\mathbf{v}$, and the weight derives from $V=mgy_P=-mg\ell\cos\theta$, so $\mathbf{W}\cdot\mathbf{v}=-dV/dt$. Hence

$$\frac{d}{dt}\Big(\tfrac12 m\big(\dot\ell^2+\ell^2\dot\theta^2\big)-mg\ell\cos\theta\Big)=Tu$$

The string is massless and the ring smooth, so the tension has the same value T at the free end, which moves with speed u along the pull: the hauler's power is Tu. The mechanical energy of P is **not** a first integral. Integrating the balance from 0 to t (Barrow's rule) gives instead

$$E_{kin}(t)+V(t)-W_{haul}(t)=E_{kin}(0)+V(0),\qquad W_{haul}(t)=\int_0^t T\,u\,dt'$$

that is, the energy of P grows by exactly the work done by the hauler. This is an energy balance, not a first integral in the strict sense: $W_{haul}(t)$ depends on the whole history $T(t')$, not on the state $(\theta,\dot\theta,t)$ alone.

Step 3. Work for $g=0$

With $T=mh^2/\ell^3$, change the variable to $\ell'=\ell_0-ut'$: $d\ell'=-u\,dt'$, so $dt'=-d\ell'/u$, and the limits become $t'=0\mapsto\ell'=\ell_0$, $t'=t\mapsto\ell'=\ell$:

$$W_{haul}=\int_0^t Tu\,dt'=\int_{\ell_0}^{\ell}\frac{mh^2}{\ell'^3}\,u\,\Big(-\frac{d\ell'}{u}\Big)=\int_{\ell}^{\ell_0}\frac{mh^2}{\ell'^3}\,d\ell'$$

An antiderivative of $mh^2\ell'^{-3}$ is $F(\ell')=-\dfrac{mh^2}{2\ell'^2}$ (check: $F'=-\frac{mh^2}{2}(-2)\ell'^{-3}=mh^2\ell'^{-3}$). Barrow's rule:

$$W_{haul} = F(\ell_0) - F(\ell) = -\frac{mh^2}{2\ell_0^2} - \Big(-\frac{mh^2}{2\ell^2}\Big) = \frac{mh^2}{2}\Big(\frac{1}{\ell^2} - \frac{1}{\ell_0^2}\Big)$$

With the data: $\frac{0.5\cdot 4}{2}\big(4 - 0.25\big) = 3.75$ J.

Step 4. Check

$E_{kin} = \frac{1}{2}m\big(u^2 + h^2/\ell^2\big)$: from $\frac{1}{2}\cdot 0.5\,(0.25 + 1)$ to $\frac{1}{2}\cdot 0.5\,(0.25 + 16)$, a change of $\frac{1}{2}\cdot 0.5 \cdot 15 = 3.75$ J. The radial part u^2 is the same at both ends, so all the work goes into the transverse motion.

KEY RESULTS

- $\mathbf{T}\cdot\mathbf{v} = -T\,\dot\ell = T\,u > 0$: the tension does positive work although it is "a constraint force".
- $W_{haul} = \dfrac{mh^2}{2}\Big(\dfrac{1}{\ell^2} - \dfrac{1}{\ell_0^2}\Big) = 3.75\ \text{J} = \Delta E_{kin}$.

Part (f) · 2 points

Extension: the amplitude of small swings while hauling

EXTENSION

This part goes beyond the course (adiabatic invariants and Bessel functions are not in the Notes). It is included because the result is a classic trap: neither the energy nor the angular momentum is conserved, so neither gives the amplitude.

For small angles, rewrite the equation of (c) with ℓ as the independent variable. Show that, when the hauling is slow ($u \ll \sqrt{g\ell}$), the angular amplitude grows like $\ell^{-3/4}$, and check it with the data of the Setup between $\ell_0 = 1$ m and $\ell = 0.25$ m. Grading: the 2.0 points are for the change of variable (chain rule) and the reduced equation in ℓ; the solution with Bessel functions and the $\ell^{-3/4}$ law are an extension, shown for completeness and not expected in the exam.

Step 1. Change of variable

For $\theta(t) = \Theta(\ell(t))$, the chain rule gives $\dot\theta = \Theta'\dot\ell = -u\Theta'$ and $\ddot\theta = \dfrac{d}{dt}(-u\Theta') = -u\Theta''\dot\ell = u^2\Theta''$. With $\sin\theta \approx \theta$ the equation of (c) becomes

$$\ell u^2\Theta'' + 2(-u)(-u\Theta') + g\Theta = 0 \implies \ell\,\Theta'' + 2\Theta' + \frac{g}{u^2}\Theta = 0$$

and since $\dfrac{d}{d\ell}\big(\ell^2\Theta'\big) = \ell^2\Theta'' + 2\ell\Theta'$, it is $\dfrac{1}{\ell}\big(\ell^2\Theta'\big)' + \dfrac{g}{u^2}\Theta = 0$.

Step 2. Exact solution

With $x = g\ell/u^2$ the equation is $x\,y'' + 2y' + y = 0$, whose solutions are $x^{-1/2}J_1(2\sqrt{x})$ and $x^{-1/2}Y_1(2\sqrt{x})$ (Bessel functions of order 1). So

$$\theta = \ell^{-1/2}\Big[A\,J_1\Big(\frac{2\sqrt{g\ell}}{u}\Big) + B\,Y_1\Big(\frac{2\sqrt{g\ell}}{u}\Big)\Big]$$

Step 3. Slow hauling

For large argument $J_1(z) \approx \sqrt{2/(\pi z)}\cos(z - 3\pi/4)$ (and Y_1 with a sine). With $z \propto \ell^{1/2}$ the envelope is $\ell^{-1/2}z^{-1/2} \propto \ell^{-3/4}$:

$$\theta_a \propto \ell^{-3/4}$$

The condition $z \gg 1$ is $u \ll \sqrt{g\ell}$: the length changes little during one swing. The same law follows from the adiabatic invariant E_{osc}/ω with $E_{osc} \approx \tfrac{1}{2}mg\ell\theta_a^2$ and $\omega = \sqrt{g/\ell}$: $\ell^{3/2}\theta_a^2$ stays constant.

Step 4. Numbers

At $\ell = 0.25$ m, $z = 2\sqrt{9.81 \cdot 0.25}/0.01 = 313 \gg 1$ and $u/\sqrt{g\ell} = 0.0064$. The law predicts $\theta_a = 0.05\,(0.25)^{-3/4} = 0.05 \cdot 2.828 = 0.1414$ rad: the swing angle almost triples, while the horizontal excursion $\ell\theta_a \propto \ell^{1/4}$ shrinks by the factor 0.707. A Runge-Kutta integration of $(\ell_0 - ut)\ddot{\theta} - 2u\dot{\theta} + g\sin\theta = 0$ gives 0.1401 rad at the last turning point ($\ell = 0.253$ m), where the law gives 0.1400 rad.

KEY RESULTS

- Graded: $\dot\theta = -u\,\theta'$, $\ddot\theta = u^2\theta''$ (chain rule) and the reduced equation below. Everything after it (Bessel solution, $\ell^{-3/4}$ law, numbers) is an extension, not expected in the exam.
- $\ell\,\theta'' + 2\theta' + \dfrac{g}{u^2}\,\theta = 0$, i.e. $\dfrac{1}{\ell}\dfrac{d}{d\ell}\big(\ell^2\theta'\big) + \dfrac{g}{u^2}\theta = 0$.
- $\theta = \ell^{-1/2}\big[A\,J_1(z) + B\,Y_1(z)\big]$, $z = 2\sqrt{g\ell}/u$; for $z \gg 1$, $J_1, Y_1 \sim \sqrt{2/(\pi z)}$ times a cosine, so the amplitude is $\propto \ell^{-1/2}\ell^{-1/4} = \ell^{-3/4}$.
- From 1 m to 0.25 m: $\theta_a = 0.05 \cdot 4^{3/4} = 0.1414$ rad; a numerical integration of the full equation agrees within $0.2\,\%$.

Interpretation and checks

- **Dimensional check:** every term of $\ell\ddot\theta + 2\dot\ell\dot\theta + g\sin\theta$ is $\mathrm{m\,s^{-2}}$; $[mh^2/\ell^3] = \mathrm{kg\,m^4 s^{-2}/m^3} = \mathrm{N}$; $[Tu] = \mathrm{W}$; $g\ell/u^2$ is dimensionless.
- **Limits:** $u = 0$ gives the simple pendulum with $T = m(g\cos\theta + \ell\dot\theta^2)$ and energy conservation; $u < 0$ (paying the string out) turns $-2u\dot\theta$ into a true damping and the amplitude decreases like $\ell^{-3/4}$.
- **Common mistakes in the exam:** (1) forgetting the Coriolis-like term $2\dot\ell\dot\theta$ when ℓ varies; (2) claiming that the tension does no work because "it is a constraint force" (it does, because the constraint is rheonomic); (3) writing energy conservation for P; (4) using $\ell^2\dot\theta = \text{const}$ with gravity (the weight has a moment about O); (5) believing that T stays bounded in the gravity-free case.